

Factoring Algebraic Expressions (Part IV)

Combined Types of Factoring

$$\textcircled{1} 6x^4 + 18x^3 + 12x^2 = 6x^2(x^2 + 3x + 2) \\ = \boxed{6x^2(x+1)(x+2)}$$

$$\textcircled{2} 12y^3 - 27y = 3y(4y^2 - 9) \\ = \boxed{3y(2y+3)(2y-3)}$$

$$\star \textcircled{3} -12x^8 - 4x^7 + 8x^6 = -4x^6(3x^2 + x - 2) \\ = \boxed{-4x^6(3x-2)(x+1)}$$

$$\textcircled{4} 9x^3 - 36x = 9x(x^2 - 4) \\ = \boxed{9x(x-2)(x+2)}$$

$$\textcircled{5} z^8 - 8z^7 + 16z^6 = z^6(z^2 - 8z + 16) \\ = \boxed{z^6(z-4)(z-4)}$$

$$\textcircled{6} 50y^4 - 128y^2 = 2y^2(25y^2 - 64) \\ = \boxed{2y^2(5y-8)(5y+8)}$$

$$\star \textcircled{7} -6z^3 + 38z^2 - 60z = -2z(3z^2 - 19z + 30) \\ = \boxed{-2z(3z-10)(z-3)}$$

$$\textcircled{8} 16y^3 - 4y = 4y(4y^2 - 1) \\ = \boxed{4y(2y-1)(2y+1)}$$

$$\textcircled{9} 3x^3 + 21x^2 + 30x = 3x(x^2 + 7x + 10) \\ = \boxed{3x(x+2)(x+5)}$$

$$\star \textcircled{10} -2x^2 + 32 = -2(x^2 - 16) \\ = \boxed{-2(x+4)(x-4)}$$

$$\textcircled{11} 4y^3 + 4y^2 - 120y = 4y(y^2 + y - 30) \\ = \boxed{4y(y+6)(y-5)}$$

$$\textcircled{12} 5y^4 + 10y^3 - 40y^2 = 5y^2(y^2 + 2y - 8) \\ = \boxed{5y^2(y+4)(y-2)}$$

$$\star \textcircled{13} -18x^6 + 18x^5 + 8x^4 = -2x^4(9x^2 - 9x - 4) \\ = \boxed{-2x^4(3x+1)(3x-4)}$$

$$\textcircled{14} 14y^5 + 49y^4 + 42y^3 = 7y^3(2y^2 + 7y + 6) \\ = \boxed{7y^3(2y+3)(y+2)}$$

$$\textcircled{15} 5z^4 - 20z^2 = 5z^2(z^2 - 4) \\ = \boxed{5z^2(z-2)(z+2)}$$

$$\textcircled{16} z^5 - 13z^4 + 40z^3 = z^3(z^2 - 13z + 40) \\ = \boxed{z^3(z-5)(z-8)}$$

$$\textcircled{17} 10y^5 - 15y^4 - 25y^3 = 5y^3(2y^2 - 3y - 5) \\ = \boxed{5y^3(2y-5)(y+1)}$$

$$\bullet \textcircled{18} 4x^6 + 16x^5 + 6x^3 + 24x^2 = 2x^2(2x^4 + 8x^3 + 3x + 12) \\ = 2x^2[2x^3(x+4) + 3(x+4)] \\ = 2x^2[(x+4)(2x^3+3)] \\ = \boxed{2x^2(x+4)(2x^3+3)}$$

$$\star \textcircled{19} -y^8 - 5y^6 + 3y^5 + 15y^3 = -y^3(y^5 + 5y^3 - 3y^2 - 15) \\ = -y^3[y^3(y^2+5) - 3(y^2+5)] \\ = -y^3[(y^2+5)(y^3-3)] \\ = \boxed{-y^3(y^2+5)(y^3-3)}$$

$$\textcircled{20} 12x^6 + 6x^5 + 18x^4 + 9x^3 = 3x^3(4x^3 + 2x^2 + 6x + 3) \\ = 3x^3[2x^2(2x+1) + 3(2x+1)] \\ = 3x^3[(2x+1)(2x^2+3)] \\ = \boxed{3x^3(2x+1)(2x^2+3)}$$

$$\star \textcircled{21} -3y^6 + 9y^5 - 12y^4 + 36y^3 = -3y^3(y^3 - 3y^2 + 4y - 12) \\ = -3y^3[y^2(y-3) + 4(y-3)] \\ = -3y^3[(y-3)(y^2+4)] \\ = \boxed{-3y^3(y-3)(y^2+4)}$$

$$\begin{aligned} \textcircled{22} -2x^5 - 3x^3 + 44x &= -x(2x^4 + 3x^2 - 44) \\ &= -x(2x^2 + 11)(x^2 - 4) \\ &= \boxed{-x(2x^2 + 11)(x - 2)(x + 2)} \end{aligned}$$

$$\begin{aligned} \textcircled{23} 4z^7 + 44z^5 + 112z^3 &= 4z^3(z^4 + 11z^2 + 28) \\ &= \boxed{4z^3(z^2 + 4)(z^2 + 7)} \end{aligned}$$

$$\begin{aligned} \textcircled{24} 5z^7 - 5z^3 &= 5z^3(z^4 - 1) \\ &= 5z^3(z^2 - 1)(z^2 + 1) \\ &= \boxed{5z^3(z - 1)(z + 1)(z^2 + 1)} \end{aligned}$$

$$\begin{aligned} \textcircled{25} 3x^6 - 48x^2 &= 3x^2(x^4 - 16) \\ &= 3x^2(x^2 - 4)(x^2 + 4) \\ &= \boxed{3x^2(x - 2)(x + 2)(x^2 + 4)} \end{aligned}$$

$$\begin{aligned} \textcircled{26} 2a^4b^5 + 16a^3b^6 + 24a^2b^7 &= 2a^2b^5(a^2 + 8ab + 12b^2) \\ &= \boxed{2a^2b^5(a + 2b)(a + 6b)} \end{aligned}$$

$$\begin{aligned} \textcircled{27} 4x^{10}y^6 - 100x^8y^8 &= 4x^8y^6(x^2 - 25y^2) \\ &= \boxed{4x^8y^6(x - 5y)(x + 5y)} \end{aligned}$$

$$\begin{aligned} \textcircled{28} -6c^7d^2 - 15c^6d^3 + 9c^5d^4 &= -3c^5d^2(2c^2 + 5cd - 3d^2) \\ &= \boxed{-3c^5d^2(2c - d)(c + 3d)} \end{aligned}$$

$$\begin{aligned} \textcircled{29} c^3d^2 - 4cd^4 &= cd^2(c^2 - 4d^2) \\ &= \boxed{cd^2(c - 2d)(c + 2d)} \end{aligned}$$

$$\begin{aligned} \textcircled{30} 12a^7b^2 + 2a^6b^3 - 4a^5b^4 &= 2a^5b^2(6a^2 + ab - 2b^2) \\ &= \boxed{2a^5b^2(3a + 2b)(2a - b)} \end{aligned}$$

$$\begin{aligned} \textcircled{31} 4c^7d^6 - 64c^5d^8 &= 4c^5d^6(c^2 - 16d^2) \\ &= \boxed{4c^5d^6(c - 4d)(c + 4d)} \end{aligned}$$

$$\begin{aligned} \textcircled{32} -x^4y^2 + 7x^3y^3 - 12x^2y^4 &= -x^2y^2(x^2 - 7xy + 12y^2) \\ &= \boxed{-x^2y^2(x - 3y)(x - 4y)} \end{aligned}$$

* ③③ $-8x^7y^2 + 18x^5y^2 = -2x^5y^2(4x^2 - 9)$
 $= -2x^5y^2(2x - 3)(2x + 3)$

③④ $2a^5b^7 + 16a^4b^8 + 30a^3b^9 = 2a^3b^7(a^2 + 8ab + 15b^2)$
 $= 2a^3b^7(a + 3b)(a + 5b)$

③⑤ $36x^{10}y^6 - 16x^8y^8 = 4x^8y^6(9x^2 - 4y^2)$
 $= 4x^8y^6(3x + 2y)(3x - 2y)$

* ③⑥ $-2x^6y^2 + 12x^5y^3 - 16x^4y^4 = -2x^4y^2(x^2 - 6xy + 8y^2)$
 $= -2x^4y^2(x - 2y)(x - 4y)$

③⑦ $25c^4d^3 - 4c^2d^5 = c^2d^3(25c^2 - 4d^2)$
 $= c^2d^3(5c - 2d)(5c + 2d)$